TM Forum Specification

TMF686 Topology Graph Discovery API User Guide

Experimental

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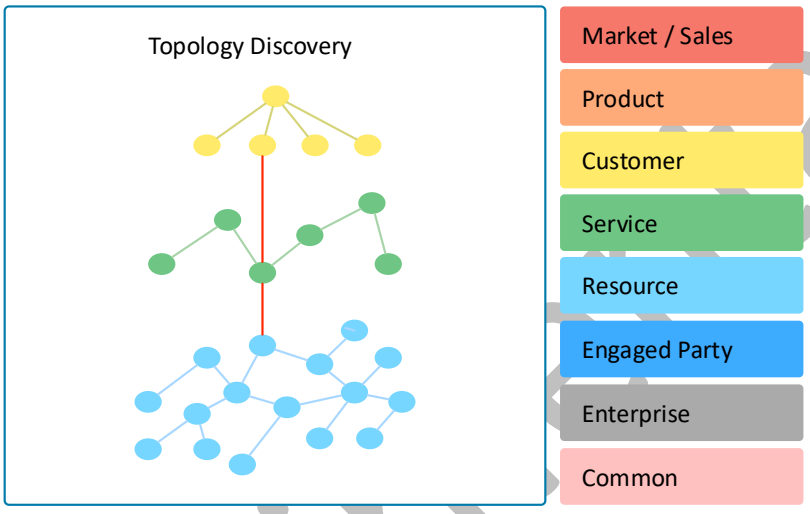
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Introduction

The following document is the user guide of the REST API for Topology Graph Discovery management. It includes the model definition as well as all available operations.

Overview

The Topology Discovery Service provides an overlay view of the relational information available across TM Forum domains in the form of directed graphs.



Roles

The Topology Graph Discovery Service may be viewed from the perspective of three actors:

Topology Consumer

The *Topology Consumer* uses this API (TMF686) to interact with the *Topology Discovery Service* for the management of topology discovery and topology graphs.

Topology Discovery Service

The *Topology Discovery Service* manages collections of topology graphs on behalf of *Topology Consumers*. Those graphs may be discovered from *Topology Providers* or created directly by *Topology Consumers*.

Topology Provider

The Topology Providers are producers of any TM Forum Open APIs such as catalogs and inventories.



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Discovery Directives

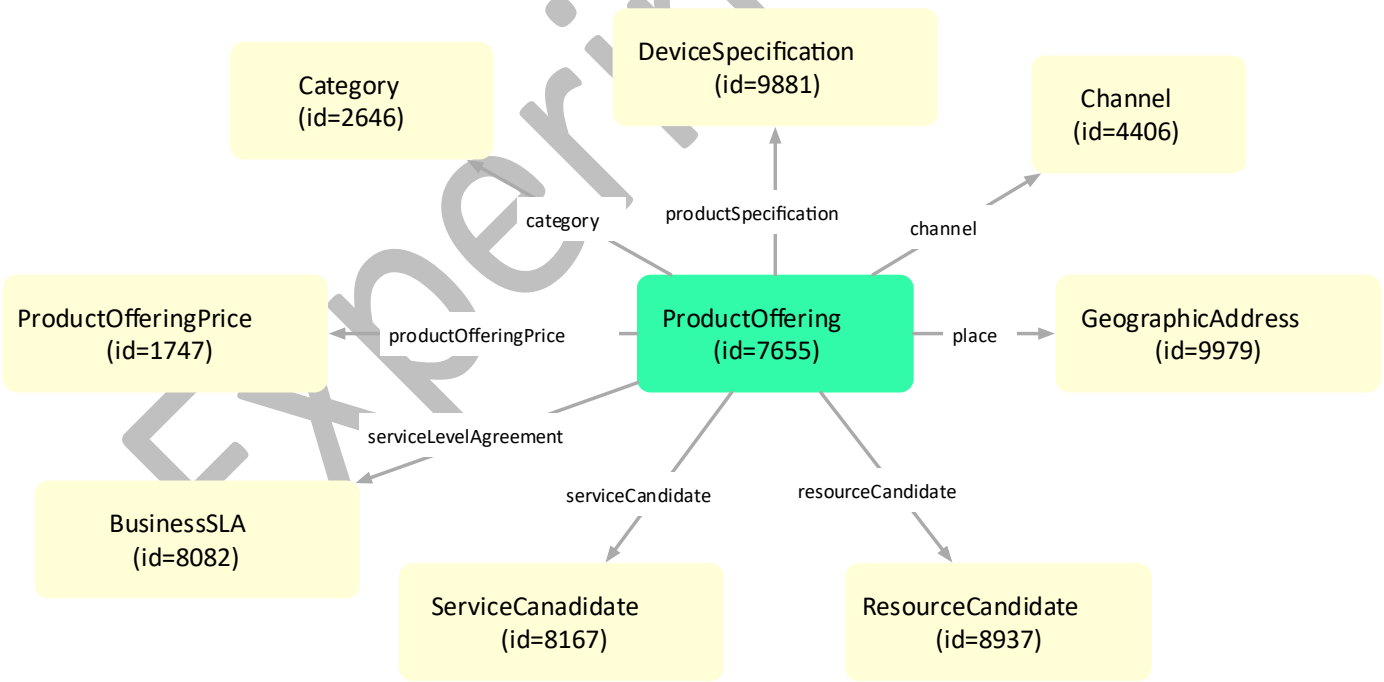
Topology Consumers may request the creation of a topology graph by representing information available in one or more Topology Providers. A URI including the scheme, host, resource path and optional query, provides the Topology Discovery Service with a starting point for discovery. If no other directives are provided the default discovery rules shall be used.

Default Discovery

In the absence of more specific discovery directives a topology graph will be created where each of the items in the response from the *Topology Provider* is represented by a vertex. Each attribute with a value having *@baseType* of *EntityRef*, or array of, shall be represented by an *Edge* and a *Vertex* shall represent the referenced entity. By default the name of an *Edge* created in this way shall have the value of it's *name* attribute equal to the attribute name which caused it while the *name* value of a *Vertex* shall be equal to that of the referenced entity.

Example

The graphical representation of a directed graph below depicts the result of topology discovery on the example Product Offering given in TMF620. The Topology Provider URI for the discovery was:
`/productCatalogManagement/v4/productOffering/7655`



This example is used in one of the topology Discovery operation samples in a following section.

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SAMPLE USE CASES

Reader will find example of use cases using Usage API in “Open Digital Business Scenarios and Use Cases” document.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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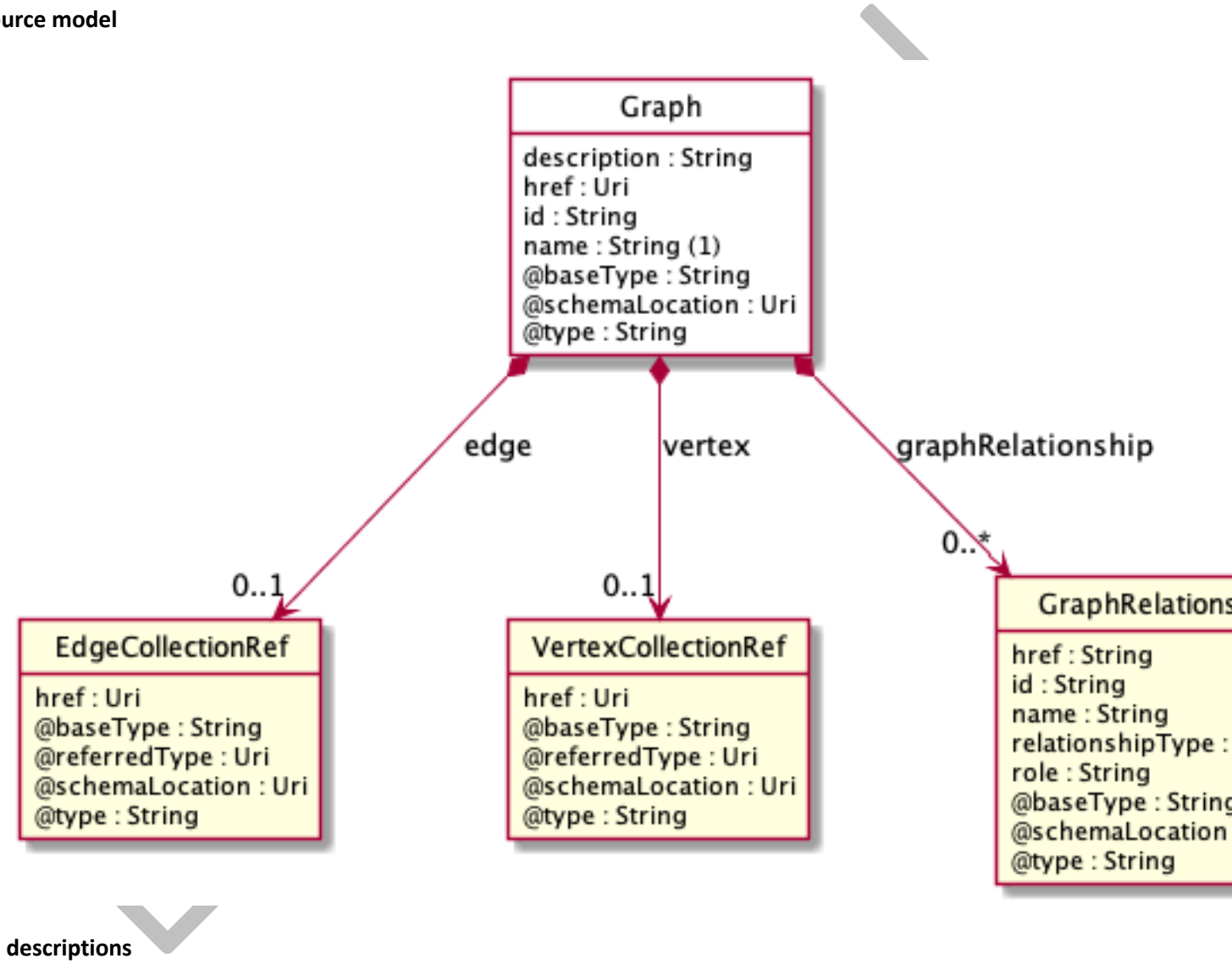
RESOURCE MODEL

Managed Entity and Task Resource Models

Graph resource

Graph is a directed graph representing edges connecting vertices.

Resource model



Field descriptions

Graph fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. Description of this REST resource.
edge	An edge collection reference (EdgeCollectionRef). Graph edge collection reference.
graphRelationship	A list of graph relationships (GraphRelationship [*]). Relationship to another Graph.
name	A string. Name given to this REST resource.
vertex	A vertex collection reference (VertexCollectionRef). Graph vertex collection reference.

GraphRelationship sub-resource

A relationship between Graphs.

href	A string. Reference of the target Graph.
id	A string. Unique identifier of target Graph.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. The name given to the target Grapph instance.
relationshipType	A string. Type of relationship such as adjacency, connectivity.
role	A string. The association role for this Graph.

EdgeCollectionRef relationship

Graph edge collection reference.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	An uri (Uri). The type of the referred entity.
href	An uri (Uri). Hyperlink reference.

VertexCollectionRef relationship

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Graph vertex collection reference.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	An uri (Uri). The type of the referred entity.
href	An uri (Uri). Hyperlink reference.

Json representation sample

We provide below the json representation of an example of a 'Graph' resource object


```
{
  "id": "b965a5b8",
  "href": "/topologyDiscovery/v4/graph/b965a5b8",
  "name": "5GC",
  "description": "5G Core Subnet",
  "@type": "Graph",
  "vertex": {
    "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex"
  },
  "edge": {
    "href": "/topologyDiscovery/v4/graph/b965a5b8/edge"
  }
}
```

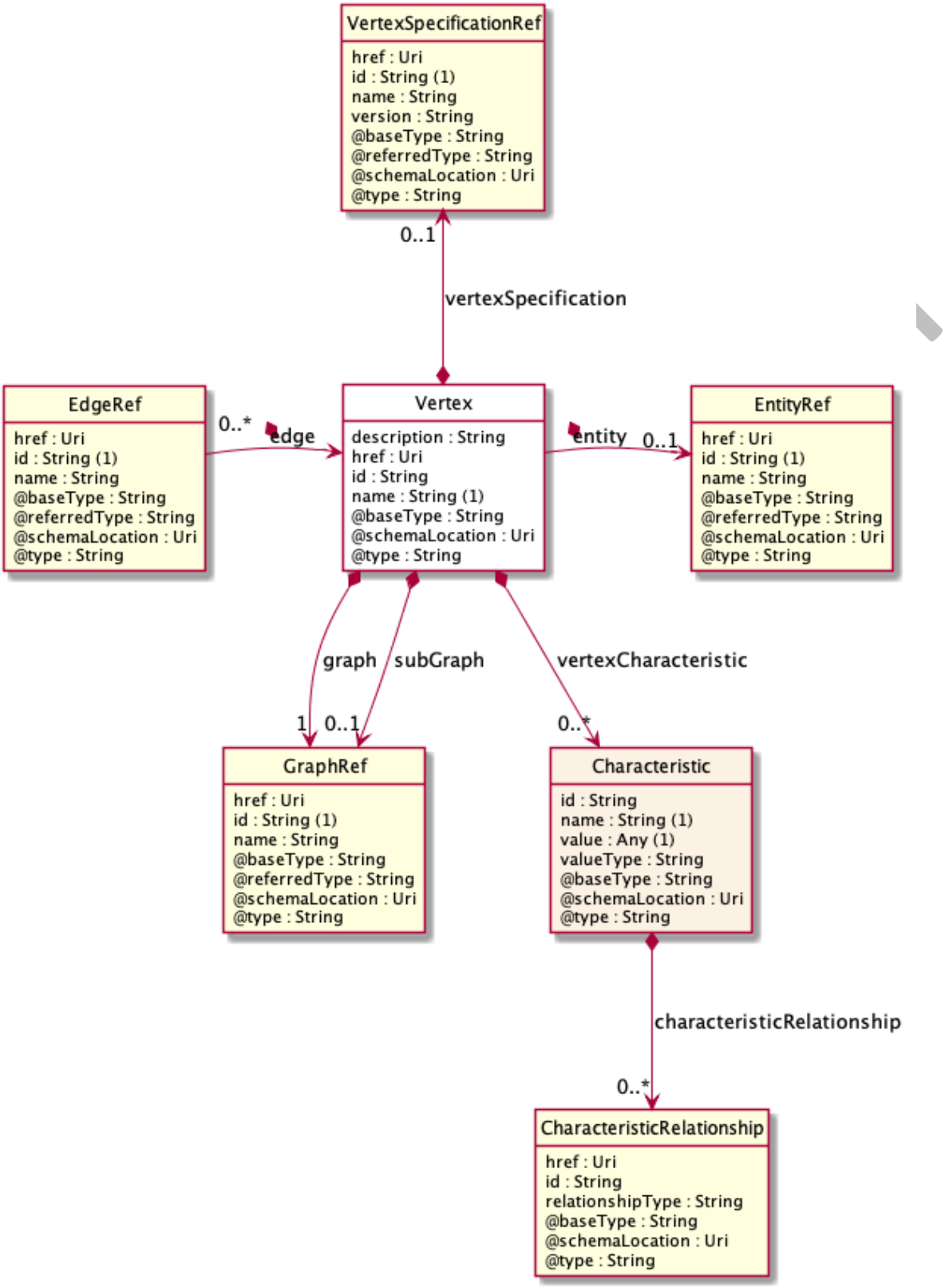

Vertex resource

A Vertex is connected by directed Edges to other vertices in a Graph.

Resource model

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Field descriptions

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Vertex fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. Description of this REST resource.
edge	A list of edge references (EdgeRef [*]). Directed Edges which this Vertex is connected by. (optional, read only).
entity	An entity reference (EntityRef). Entity which this Vertex represents.
graph	A graph reference (GraphRef). Graph which this Vertex is contained in.
name	A string. Name given to this REST resource.
subGraph	A graph reference (GraphRef). SubGraph which this Vertex represents.
vertexCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
vertexSpecification	A vertex specification reference (VertexSpecificationRef). A Vertex is connected by directed Edges to other vertices in a Graph. A VertexSpecification is a base class that represents a generic means for implementing a particular type of Vertex. In essence, a VertexSpecification defines the common attributes and relationships of a set of related Vertices, while Vertex defines a specific instance that is based on a particular VertexSpecification.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.

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id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

EdgeRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

EntityRef relationship

Entity reference schema to be use for all entityRef class.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

GraphRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

VertexSpecificationRef relationship

A Vertex is connected by directed Edges to other vertices in a Graph.
A VertexSpecification is a base class that represents a generic means for implementing a particular type of Vertex. In essence, a VertexSpecification defines the common attributes and relationships of a set of related Vertices, while Vertex defines a specific instance that is based on a particular VertexSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Vertex Specification version.

Json representation sample

We provide below the json representation of an example of a 'Vertex' resource object

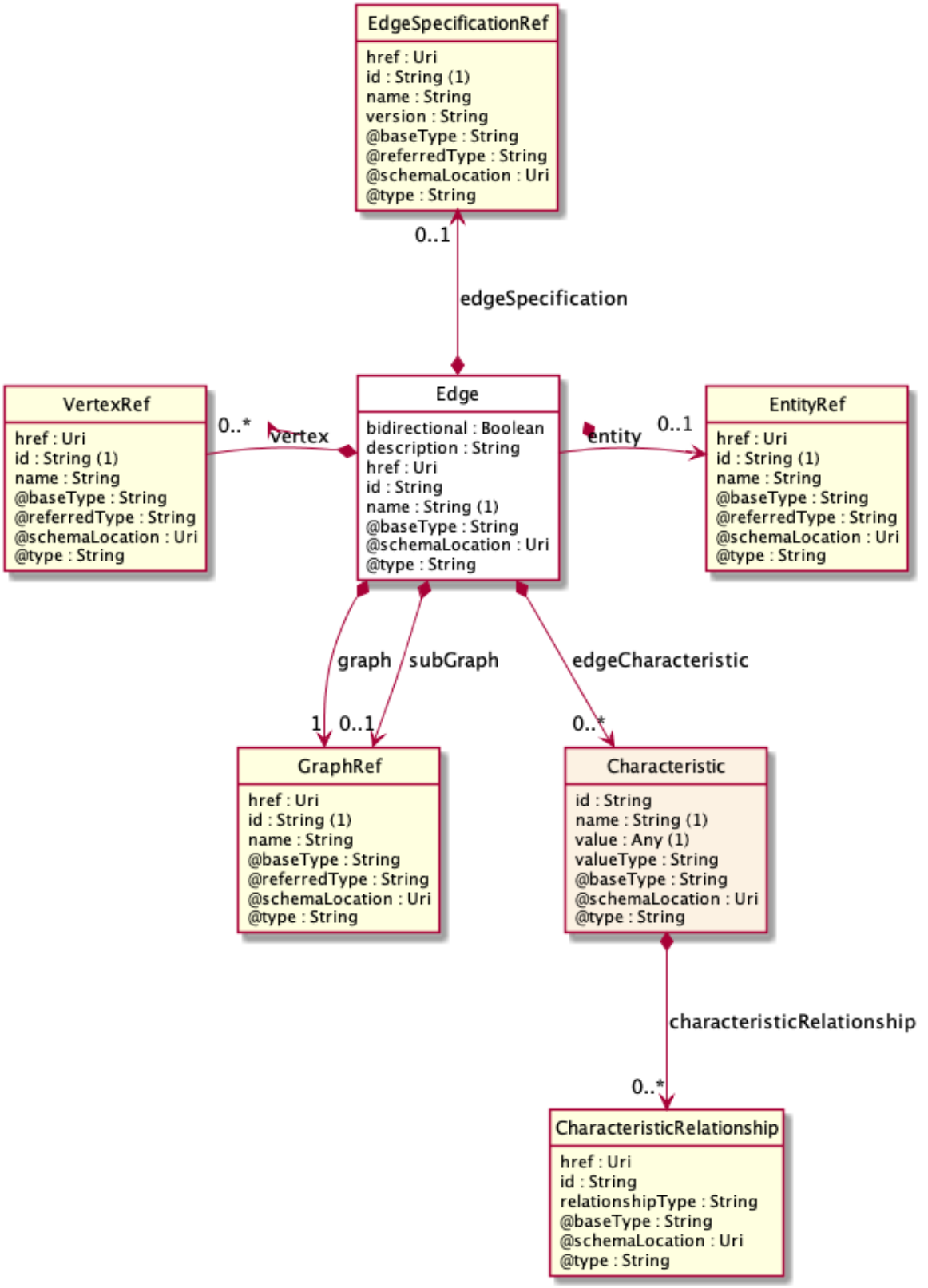

```
{
  "id": "8e516fb0",
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
  "name": "AMF",
  "description": "5G Core AMF",
  "@type": "Vertex",
  "vertexSpecification": {
    "id": "6106de14",
    "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
    "name": "VNF",
    "@referredType": "VertexSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edge": [
    {
      "id": "446d7249",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
      "name": "N8",
      "@referredType": "Edge"
    }
  ],
  "vertexCharacteristic": [
    {
      "name": "datacenter",
      "value": "DC005"
    }
  ],
  "entity": {
    "id": "5e86ec0a",
    "href": "/resourceInventory/v4/resource/5e86ec0a",
    "name": "NetworkSliceSubnet=1,ManagedElement=1,AMFFunction=1",
    "@referredType": "ResourceFunction"
  }
}
```


Edge resource

Edge is a directed edge connecting vertices in a directed Graph.

Resource model

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Field descriptions

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Edge fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
bidirectional	A boolean. Indicates if this Edge is birectional (true) or unidirection originating from the first vertex (false).
description	A string. Description of this REST resource.
edgeCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
edgeSpecification	An edge specification reference (EdgeSpecificationRef). A directed Edge connects Vertices in a Graph. An EdgeSpecification is a base class that represents a generic means for implementing a particular type of Edge. In essence, an EdgeSpecification defines the common attributes and relationships of a set of related Edges, while Edge defines a specific instance that is based on a particular EdgeSpecification.
entity	An entity reference (EntityRef). Entity which this Edge represents.
graph	A graph reference (GraphRef). Graph which this Edge is contained in.
name	A string. Name given to this REST resource.
subGraph	A graph reference (GraphRef). SubGraph which this Edge represents.
vertex	A list of vertex references (VertexRef [*]). Graph Vertices connected by this Edge.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

EdgeSpecificationRef relationship

A directed Edge connects Vertices in a Graph.
An EdgeSpecification is a base class that represents a generic means for implementing a particular type of Edge. In essence, an EdgeSpecification defines the common attributes and relationships of a set of related Edges, while Edge defines a specific instance that is based on a particular EdgeSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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version	A string. Edge Specification version.

EntityRef relationship

Entity reference schema to be use for all entityRef class.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

GraphRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

VertexRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'Edge' resource object


```
{
  "id": "446d7249",
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
  "name": "N8",
  "description": "5G Core N8 Reference Interface",
  "@type": "Edge",
  "edgeSpecification": {
    "id": "95e66eb7",
    "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
    "name": "Transport",
    "@referredType": "EdgeSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edgeCharacteristic": [
    {
      "name": "provider",
      "value": "SDN45"
    }
  ],
  "bidirectional": true,
  "vertex": [
    {
      "id": "8e516fb0",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
      "name": "AMF",
      "@referredType": "Vertex"
    },
    {
      "id": "9b9f1c66",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/9b9f1c66",
      "name": "UDM",
      "@referredType": "Vertex"
    }
  ],
  "entity": {
    "id": "1fbfac41",
    "href": "/resourceInventory/v4/resource/1fbfac41",
    "name": "NetworkSlice=1,NetworkSliceSubnet=1,ManagedElement=1,Link_AMF_UDM=1",
```


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```
      "@referredType": "ResourceFunction"
    }
}
```

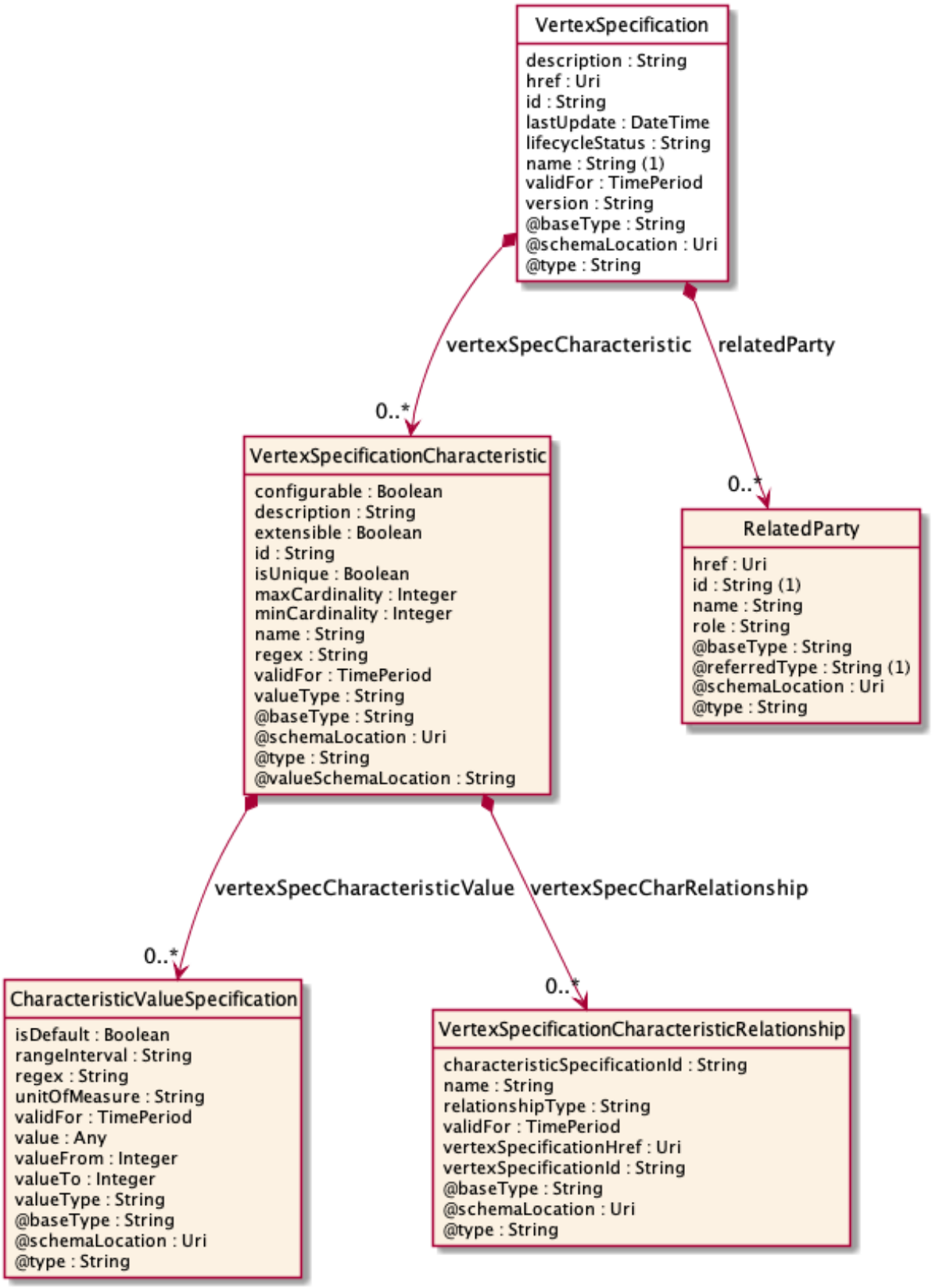
Vertex Specification resource

A Vertex is connected by directed Edges to other vertices in a Graph.
A VertexSpecification is a base class that represents a generic means for implementing a particular type of Vertex. In essence, a VertexSpecification defines the common attributes and relationships of a set of related Vertices, while Vertex defines a specific instance that is based on a particular VertexSpecification.

Resource model

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Field descriptions

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VertexSpecification fields

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
description	A string. Description of this REST resource.
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the vertex specification.
name	A string. Name given to this REST resource.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
validFor	A time period. The period for which this REST resource is valid.
version	A string. Vertex Specification version.
vertexSpecCharacteristic	A list of vertex specification characteristics (VertexSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a VertexSpecification.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.

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unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

VertexSpecificationCharacteristic sub-resource

This class defines the characteristic features of a vertex specification. Every VertexSpecification has a variety of important attributes, methods, constraints, and relationships, which distinguish a vertex specification from other vertex specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.

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description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristc. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
vertexSpecCharRelationship	A list of vertex specification characteristic relationships (VertexSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
vertexSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a VertexSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a

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	corresponding VertexSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

VertexSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among VertexSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A time period. The period for which the object is valid.
vertexSpecificationHref	An uri (Uri). Hyperlink reference to the vertex specification containing the target characteristic.
vertexSpecificationId	A string. Unique identifier of the vertex specification containing the target characteristic.

Json representation sample

We provide below the json representation of an example of a 'VertexSpecification' resource object


```
{
  "id": "6106de14",
  "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
  "@type": "VertexSpecification",
  "name": "VNF",
  "description": "Virtual Network Function (VNF)",
  "lastUpdate": "2021-04-08T12:22:08Z",
  "lifecycleStatus": "Active",
  "version": "1",
  "vertexSpecCharacteristic": [
    {
      "name": "datacenter",
      "description": "Identifies the datacenter where the VNF is located",
      "valueType": "String",
      "configurable": true
    }
```


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```
    ]
}
```

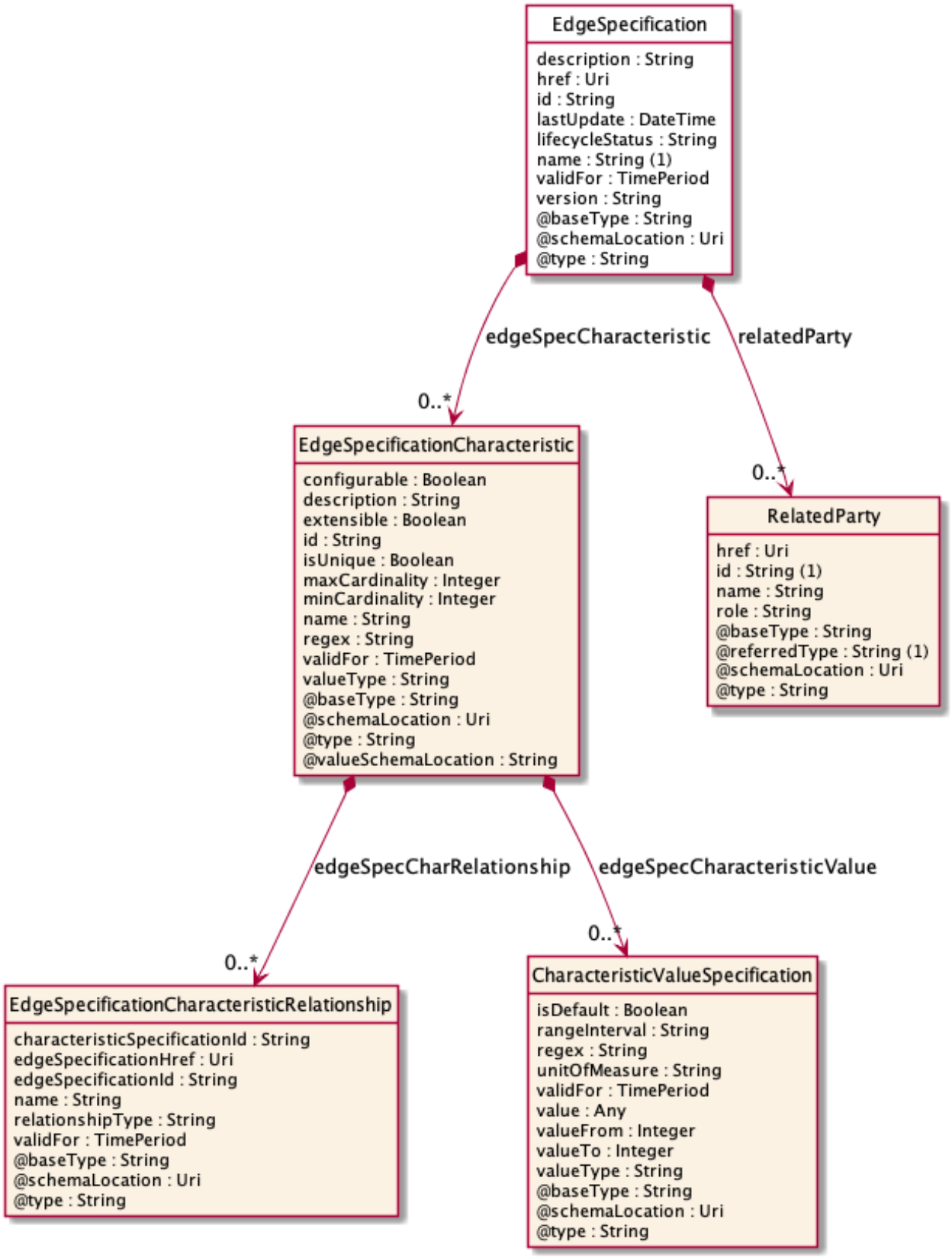
Edge Specification resource

A directed Edge connects Vertices in a Graph.
An EdgeSpecification is a base class that represents a generic means for implementing a particular type of Edge. In essence, an EdgeSpecification defines the common attributes and relationships of a set of related Edges, while Edge defines a specific instance that is based on a particular EdgeSpecification.

Resource model

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Field descriptions

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EdgeSpecification fields

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
description	A string. Description of this REST resource.
edgeSpecCharacteristic	A list of edge specification characteristics (EdgeSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of an EdgeSpecification.
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the edge specification.
name	A string. Name given to this REST resource.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
validFor	A time period. The period for which this REST resource is valid.
version	A string. Edge Specification version.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.

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unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

EdgeSpecificationCharacteristic sub-resource

This class defines the characteristic features of an edge specification. Every EdgeSpecification has a variety of important attributes, methods, constraints, and relationships, which distinguish an edge specification from other edge specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristc. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.

Experimental

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name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
edgeSpecCharRelationship	A list of edge specification characteristic relationships (EdgeSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
edgeSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a EdgeSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding EdgeSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

EdgeSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among EdgeSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
edgeSpecificationHref	An uri (Uri). Hyperlink reference to the edge specification containing the target characteristic.

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edgeSpecificationId	A string. Unique identifier of the edge specification containing the target characteristic.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A time period. The period for which the object is valid.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

Json representation sample

We provide below the json representation of an example of a 'EdgeSpecification' resource object


```
{
  "id": "95e66eb7",
  "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
  "@type": "EdgeSpecification",
  "name": "Transport",
  "description": "Link transport domain",
  "lastUpdate": "2021-04-08T13:54:32Z",
  "lifecycleStatus": "Active",
  "version": "1",
  "edgeSpecCharacteristic": [
    {
      "name": "provider",
      "description": "Identifies the transport provider of the link",
      "valueType": "String",
      "configurable": true
    }
  ]
```


Experimental

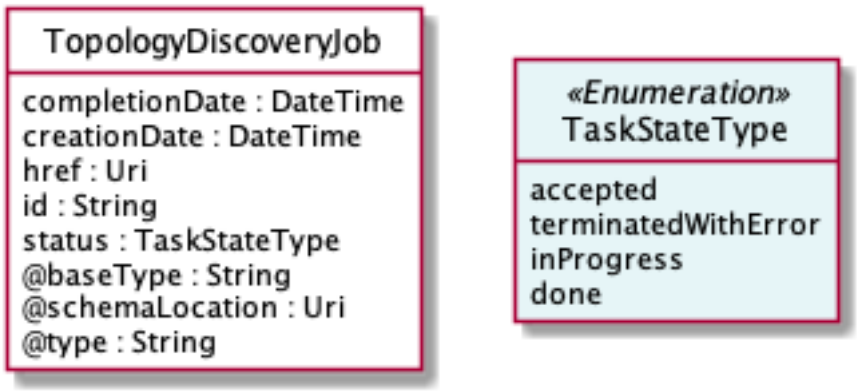
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```
}
```

Topology Discovery Job resource

Represents a task used to discover topology from providers.

Resource model



Field descriptions

TopologyDiscoveryJob fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
completionDate	A date time (DateTime). Date at which the job was completed.
creationDate	A date time (DateTime). Date at which the job was created.
status	A task state type (TaskStateType). Status of the topology discover job.

Json representation sample

We provide below the json representation of an example of a 'TopologyDiscoveryJob' resource object

```
{
  "id": "b32a6d3b",
  "href": "/topologyDiscovery/v4/topologyDiscoveryJob/b32a6d3b",
  "@type": "TopologyDiscoveryJob",
  "creationDate": "2021-04-15T14:17:04.908Z",
  "completionDate": "2021-04-15T14:17:11.845Z",
```

Experimental

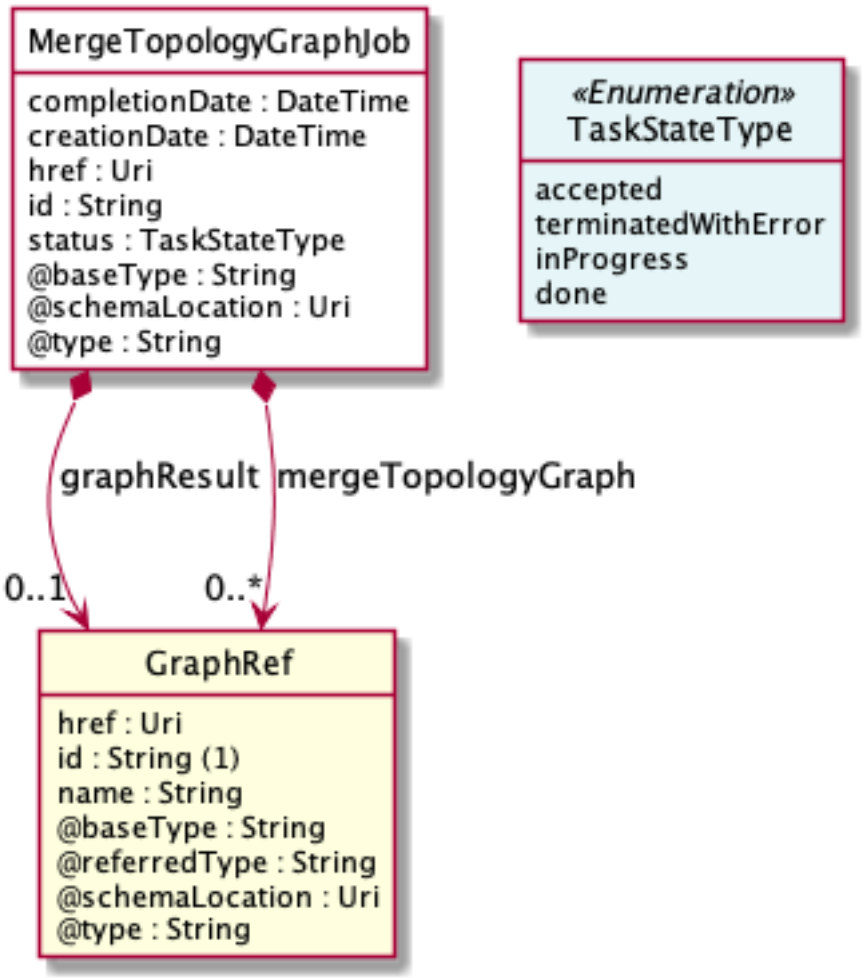
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```
  "status": "done",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ],
  "graphTemplate": {
    "name": "5GC",
    "description": "5G Core Subnet",
    "contained": true
  },
  "graphResult": {
    "id": "41fbbd95",
    "href": "/topologyDiscovery/v4/graph/41fbbd95",
    "@referredType": "Graph"
  }
}
```


Merge Topology Graph Job resource

Merges two or more Graphs into a single graph.

Resource model



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Field descriptions

MergeTopologyGraphJob fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
completionDate	A date time (DateTime). Date at which the job was completed.
creationDate	A date time (DateTime). Date at which the job was created.
graphResult	A graph reference (GraphRef). The created topology graph.
mergeTopologyGraph	A list of graph references (GraphRef [*]). URI with query string of the Topology GraphIds to be merged.
status	A task state type (TaskStateType). Status of the merge job.

GraphRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'MergeTopologyGraphJob' resource object


```
{
  "id": "333c231a",
  "href": "/topologyDiscovery/v4/MergeTopologyGraphJob/333c231a",
```


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```
  "@type": "MergeTopologyGraphJob",
  "creationDate": "2021-04-15T14:37:18.762Z",
  "completionDate": "2021-04-15T14:37:19.114Z",
  "status": "done",
  "mergeTopologyGraph": [
    {
      "id": "91f9e74a",
      "href": "/topologyDiscovery/v4/graph/91f9e74a",
      "@referredType": "Graph"
    },
    {
      "id": "0319108c",
      "href": "/topologyDiscovery/v4/graph/0319108c",
      "@referredType": "Graph"
    }
  ],
  "graph": {
    "id": "ded3c8be",
    "href": "/topologyDiscovery/v4/graph/ded3c8be",
    "@referredType": "Graph"
  }
}
```

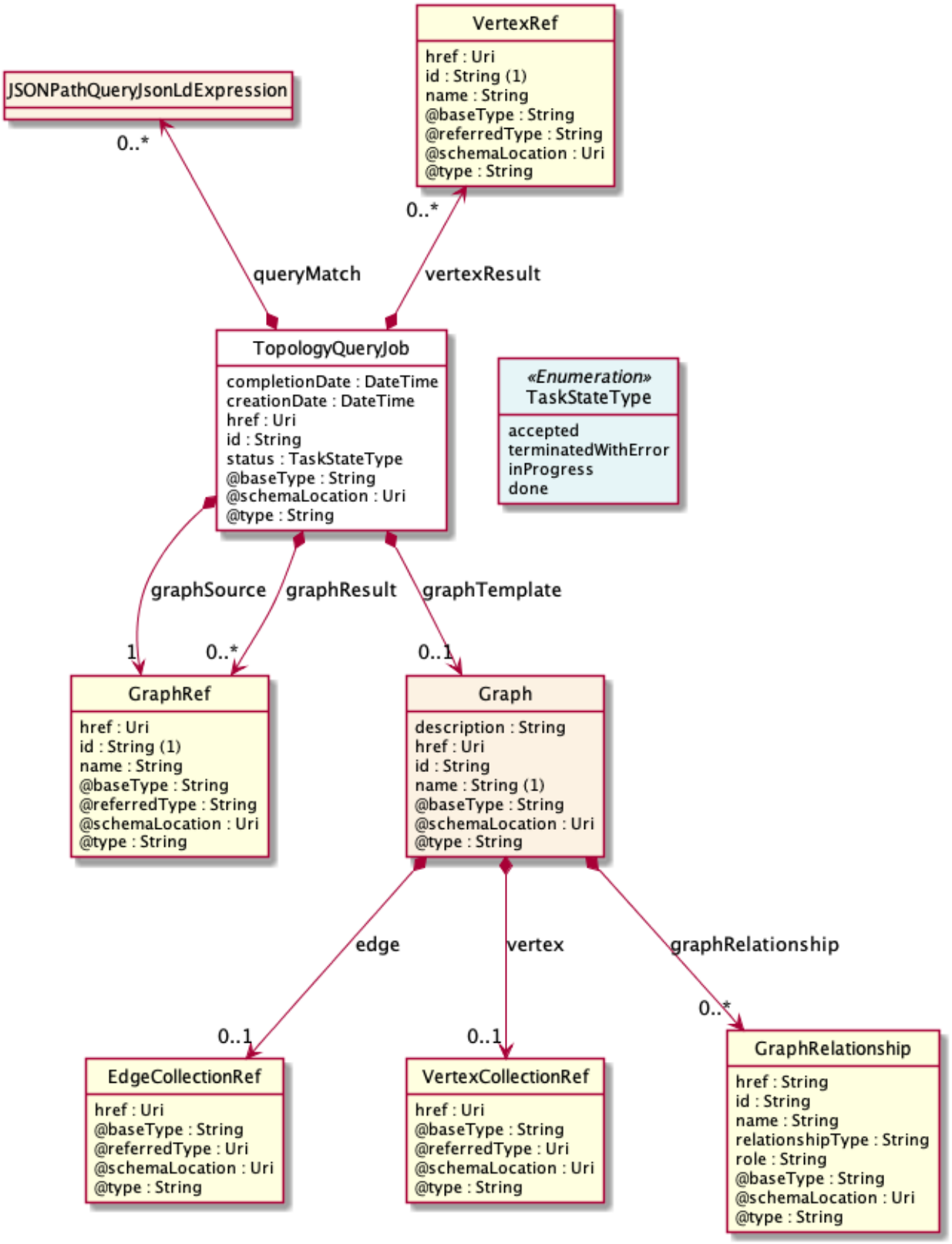

Topology Query Job resource

Represents a task used to query topology in the service.

Resource model

Experime

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Field descriptions

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TopologyQueryJob fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
completionDate	A date time (DateTime). Date at which the job was completed.
creationDate	A date time (DateTime). Date at which the job was created.
graphResult	A list of graph references (GraphRef [*]). The topology graph(s) created by the query.
graphSource	A graph reference (GraphRef). The topology graph to be queried.
graphTemplate	A graph (Graph). Attributes of the created Graph.
queryMatch	A list of j s o n path query json ld expressions (JSONPathQueryJsonLdExpression [*]). Pattern to search for in graph.
status	A task state type (TaskStateType). Status of the topology query job.
vertexResult	A list of vertex references (VertexRef [*]). The vertices result of the query.

Graph sub-resource

Graph is a directed graph representing edges connecting vertices.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. Description of this REST resource.
edge	An edge collection reference (EdgeCollectionRef). Graph edge collection reference.
graphRelationship	A list of graph relationships (GraphRelationship [*]). Relationship to another Graph.

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name	A string. Name given to this REST resource.
vertex	A vertex collection reference (VertexCollectionRef). Graph vertex collection reference.

GraphRelationship sub-resource

A relationship between Graphs.

href	A string. Reference of the target Graph.
id	A string. Unique identifier of target Graph.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. The name given to the target Grapph instance.
relationshipType	A string. Type of relationship such as adjacency, connectivity.
role	A string. The association role for this Graph.

EdgeCollectionRef relationship

Graph edge collection reference.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	An uri (Uri). The type of the referred entity.
href	An uri (Uri). Hyperlink reference.

GraphRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

VertexCollectionRef relationship

Graph vertex collection reference.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	An uri (Uri). The type of the referred entity.
href	An uri (Uri). Hyperlink reference.

VertexRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'TopologyQueryJob' resource object


```
{
  "id": "714733ce",
  "href": "/topologyDiscovery/v4/topologyQueryJob/714733ce",
  "@type": "TopologyQueryJob",
  "creationDate": "2021-04-16T33:21:54.776Z",
  "completionDate": "2021-04-16T33:21:55.245Z",
  "status": "done",
```


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```
  "query": "TUFUQ0ggKG4pClJFVFVSTiBu",
  "graphTemplate": {
    "description": "Query result subgraph",
    "contained": false
  },
  "graphResult": {
    "id": "c765bfa9",
    "href": "/topologyDiscovery/v4/graph/c765bfa9",
    "@referredType": "Graph"
  }
}
```


Notification Resource Models

21 notifications are defined for this API

Notifications related to Graph:
- GraphCreateEvent
- GraphChangeEvent
- GraphDeleteEvent

Notifications related to Vertex:
- VertexCreateEvent
- VertexChangeEvent
- VertexDeleteEvent

Notifications related to Edge:
- EdgeCreateEvent
- EdgeChangeEvent
- EdgeDeleteEvent

Notifications related to VertexSpecification:
- VertexSpecificationCreateEvent
- VertexSpecificationChangeEvent
- VertexSpecificationDeleteEvent

Notifications related to EdgeSpecification:
- EdgeSpecificationCreateEvent
- EdgeSpecificationChangeEvent
- EdgeSpecificationDeleteEvent

Notifications related to TopologyDiscoveryJob:
- TopologyDiscoveryJobCreateEvent
- TopologyDiscoveryJobStateChangeEvent

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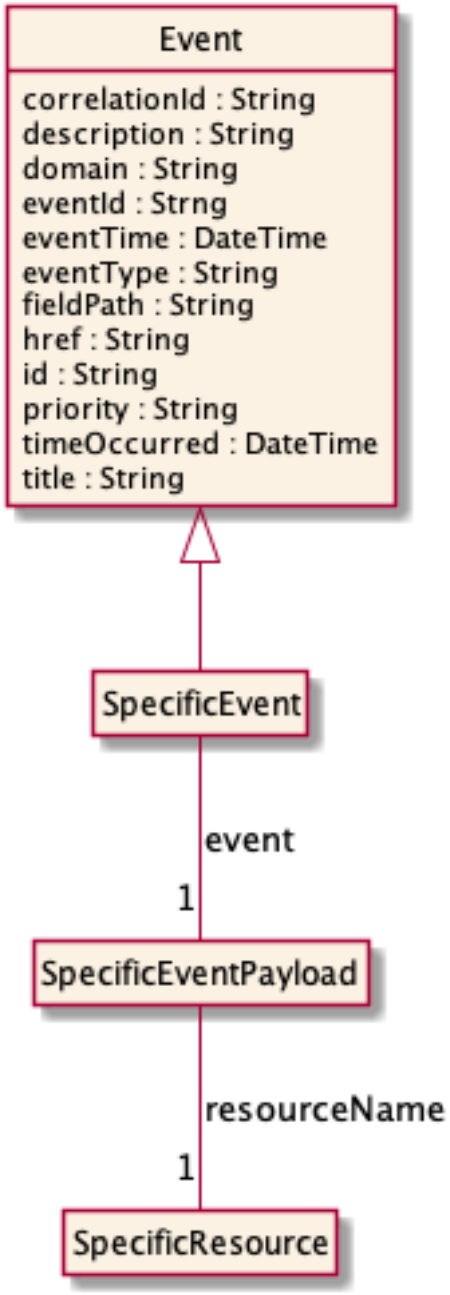
Notifications related to MergeTopologyGraphJob:
- MergeTopologyGraphJobCreateEvent
- MergeTopologyGraphJobStateChangeEvent

Notifications related to TopologyQueryJob:
- TopologyQueryJobCreateEvent
- TopologyQueryJobStateChangeEvent

The notification structure for all notifications in this API follow the pattern depicted by the figure below.
A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType).
This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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Experimental

Graph Create Event

Notification GraphCreateEvent case for resource Graph

Json representation sample

We provide below the json representation of an example of a 'GraphCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"GraphCreateEvent",
    "event": {
```

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```
    "graph" :
      {-- SEE Graph RESOURCE SAMPLE --}
  }
}
```

Graph Change Event

Notification GraphChangeEvent case for resource Graph

Json representation sample

We provide below the json representation of an example of a 'GraphChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"GraphChangeEvent",
   "event": {
    "graph" :
      {-- SEE Graph RESOURCE SAMPLE --}
  }
}
```

Graph Delete Event

Notification GraphDeleteEvent case for resource Graph

Json representation sample

We provide below the json representation of an example of a 'GraphDeleteEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"GraphDeleteEvent",
   "event": {
    "graph" :
      {-- SEE Graph RESOURCE SAMPLE --}
  }
}
```

Vertex Create Event

Notification VertexCreateEvent case for resource Vertex

Json representation sample

We provide below the json representation of an example of a 'VertexCreateEvent' notification event object

```
{
```

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```
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexCreateEvent",
   "event": {
     "vertex" :
        {-- SEE Vertex RESOURCE SAMPLE --}
  }
}
```

Vertex Change Event

Notification VertexChangeEvent case for resource Vertex

Json representation sample

We provide below the json representation of an example of a 'VertexChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexChangeEvent",
   "event": {
     "vertex" :
        {-- SEE Vertex RESOURCE SAMPLE --}
  }
}
```

Vertex Delete Event

Notification VertexDeleteEvent case for resource Vertex

Json representation sample

We provide below the json representation of an example of a 'VertexDeleteEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexDeleteEvent",
   "event": {
     "vertex" :
        {-- SEE Vertex RESOURCE SAMPLE --}
  }
}
```

Edge Create Event

Notification EdgeCreateEvent case for resource Edge

Json representation sample

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We provide below the json representation of an example of a 'EdgeCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeCreateEvent",
   "event": {
     "edge" :
       {-- SEE Edge RESOURCE SAMPLE --}
  }
}
```

Edge Change Event

Notification EdgeChangeEvent case for resource Edge

Json representation sample

We provide below the json representation of an example of a 'EdgeChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeChangeEvent",
   "event": {
     "edge" :
       {-- SEE Edge RESOURCE SAMPLE --}
  }
}
```

Edge Delete Event

Notification EdgeDeleteEvent case for resource Edge

Json representation sample

We provide below the json representation of an example of a 'EdgeDeleteEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeDeleteEvent",
   "event": {
     "edge" :
       {-- SEE Edge RESOURCE SAMPLE --}
  }
}
```

Vertex Specification Create Event

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Notification VertexSpecificationCreateEvent case for resource VertexSpecification

Json representation sample

We provide below the json representation of an example of a 'VertexSpecificationCreateEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexSpecificationCreateEvent",
   "event": {
     "vertexSpecification" :
       {-- SEE VertexSpecification RESOURCE SAMPLE --}
  }
}
```


Vertex Specification Change Event

Notification VertexSpecificationChangeEvent case for resource VertexSpecification

Json representation sample

We provide below the json representation of an example of a 'VertexSpecificationChangeEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexSpecificationChangeEvent",
   "event": {
     "vertexSpecification" :
       {-- SEE VertexSpecification RESOURCE SAMPLE --}
  }
}
```


Vertex Specification Delete Event

Notification VertexSpecificationDeleteEvent case for resource VertexSpecification

Json representation sample

We provide below the json representation of an example of a 'VertexSpecificationDeleteEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"VertexSpecificationDeleteEvent",
   "event": {
```


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```
    "vertexSpecification" :
      {-- SEE VertexSpecification RESOURCE SAMPLE --}
  }
}
```

Edge Specification Create Event

Notification EdgeSpecificationCreateEvent case for resource EdgeSpecification

Json representation sample

We provide below the json representation of an example of a 'EdgeSpecificationCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeSpecificationCreateEvent",
   "event": {
     "edgeSpecification" :
       {-- SEE EdgeSpecification RESOURCE SAMPLE --}
  }
}
```

Edge Specification Change Event

Notification EdgeSpecificationChangeEvent case for resource EdgeSpecification

Json representation sample

We provide below the json representation of an example of a 'EdgeSpecificationChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeSpecificationChangeEvent",
   "event": {
     "edgeSpecification" :
       {-- SEE EdgeSpecification RESOURCE SAMPLE --}
  }
}
```

Edge Specification Delete Event

Notification EdgeSpecificationDeleteEvent case for resource EdgeSpecification

Json representation sample

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We provide below the json representation of an example of a 'EdgeSpecificationDeleteEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"EdgeSpecificationDeleteEvent",
   "event": {
     "edgeSpecification" :
        {-- SEE EdgeSpecification RESOURCE SAMPLE --}
  }
}
```


Topology Discovery Job Create Event

Notification TopologyDiscoveryJobCreateEvent case for resource TopologyDiscoveryJob

Json representation sample

We provide below the json representation of an example of a 'TopologyDiscoveryJobCreateEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"TopologyDiscoveryJobCreateEvent",
   "event": {
     "topologyDiscoveryJob" :
        {-- SEE TopologyDiscoveryJob RESOURCE SAMPLE --}
  }
}
```


Topology Discovery Job State Change Event

Notification TopologyDiscoveryJobStateChangeEvent case for resource TopologyDiscoveryJob

Json representation sample

We provide below the json representation of an example of a 'TopologyDiscoveryJobStateChangeEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"TopologyDiscoveryJobStateChangeEvent",
   "event": {
     "topologyDiscoveryJob" :
        {-- SEE TopologyDiscoveryJob RESOURCE SAMPLE --}
  }
}
```


Exposed

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Merge Topology Graph Job Create Event

Notification MergeTopologyGraphJobCreateEvent case for resource MergeTopologyGraphJob

Json representation sample

We provide below the json representation of an example of a 'MergeTopologyGraphJobCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"MergeTopologyGraphJobCreateEvent",
   "event": {
     "mergeTopologyGraphJob" :
       {-- SEE MergeTopologyGraphJob RESOURCE SAMPLE --}
  }
}
```

Merge Topology Graph Job State Change Event

Notification MergeTopologyGraphJobStateChangeEvent case for resource MergeTopologyGraphJob

Json representation sample

We provide below the json representation of an example of a 'MergeTopologyGraphJobStateChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"MergeTopologyGraphJobStateChangeEvent",
   "event": {
     "mergeTopologyGraphJob" :
       {-- SEE MergeTopologyGraphJob RESOURCE SAMPLE --}
  }
}
```

Topology Query Job Create Event

Notification TopologyQueryJobCreateEvent case for resource TopologyQueryJob

Json representation sample

We provide below the json representation of an example of a 'TopologyQueryJobCreateEvent' notification event object

```
{
```

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```
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"TopologyQueryJobCreateEvent",
   "event": {
     "topologyQueryJob" :
       {-- SEE TopologyQueryJob RESOURCE SAMPLE --}
  }
}
```

Topology Query Job State Change Event

Notification TopologyQueryJobStateChangeEvent case for resource TopologyQueryJob

Json representation sample

We provide below the json representation of an example of a 'TopologyQueryJobStateChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"TopologyQueryJobStateChangeEvent",
   "event": {
     "topologyQueryJob" :
       {-- SEE TopologyQueryJob RESOURCE SAMPLE --}
  }
}
```

Experimental

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Complete Update of an Entity	PUT Resource	PUT must be used to completely update a resource identified by its resource URI
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Experimental

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Graph

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List graphs

GET /graph?fields=...&{filtering}

Description

This operation list graph entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get the Graph collection:

Request

```
GET /tmf-api/topologyDiscovery/v4/graph
Accept: application/json
```

Response

200

```
[
  {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "description": "5G Core Subnet",
    "@type": "Graph",
    "vertex": {
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex"
    },
    "edge": {
      "href": "/topologyDiscovery/v4/graph/b965a5b8/edge"
    }
  }
]
```

Retrieve graph

GET /graph/{id}?fields=...&{filtering}

Description

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This operation retrieves a graph entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a Graph resource.

Request

```
GET /tmf-api/topologyDiscovery/v4/graph/b965a5b8
Accept: application/json
```

Response

```
200

{
   "id": "b965a5b8",
   "href": "/topologyDiscovery/v4/graph/b965a5b8",
   "name": "5GC",
   "description": "5G Core Subnet",
   "@type": "Graph",
   "vertex": {
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex"
   },
   "edge": {
      "href": "/topologyDiscovery/v4/graph/b965a5b8/edge"
   }
}
```

Create graph

POST /graph

Description

This operation creates a graph entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Graph, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

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Mandatory Attributes	Rule
name	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
description	
edge	
graphRelationship	
vertex	

Usage Samples

Here's an example of a request for creating a Graph resource. In this example the request only passes mandatory attributes.

Request

```
POST /tmf-api/topologyDiscovery/v4/graph
Content-Type: application/json

{
   "name": "5GC"
}
```

Response

```
201

{
   "id": "b965a5b8",
   "href": "/topologyDiscovery/v4/graph/b965a5b8",
   "name": "5GC"
}
```

Patch graph

`PATCH /graph/{id}`

Description

This operation allows partial updates of a graph entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

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Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
href	
id	
@baseType	
@schemaLocation	
@type	
description	
edge	
graphRelationship	
name	
vertex	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a Graph resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/graph/b965a5b8
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "b965a5b8",
   "href": "/topologyDiscovery/v4/graph/b965a5b8",
   "name": "new name",
```

Experimental

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```
  "description": "5G Core Subnet",
  "@type": "Graph",
  "vertex": {
    "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex"
  },
  "edge": {
    "href": "/topologyDiscovery/v4/graph/b965a5b8/edge"
  }
}
```


Delete graph

`DELETE /graph/{id}`

Description

This operation deletes a graph entity.

Usage Samples

Here's an example of a request for deleting a Graph resource.

Request
DELETE /tmf-api/topologyDiscovery/v4/graph/42
Response
204

Operations on Vertex

List vertexs

`GET graph/{graphId}/vertex?fields=...&{filtering}`

Description

This operation list vertex entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Experimental

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Usage Samples

Here's an example of a request for retrieving Vertex resources.

Request

```
GET /tmf-api/topologyDiscovery/v4/vertex
Accept: application/json
```

Response

```
200

[
{
  "id": "8e516fb0",
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
  "name": "AMF",
  "description": "5G Core AMF",
  "@type": "Vertex",
  "vertexSpecification": {
    "id": "6106de14",
    "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
    "name": "VNF",
    "@referredType": "VertexSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edge": [
    {
      "id": "446d7249",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
      "name": "N8",
      "@referredType": "Edge"
    }
  ],
  "vertexCharacteristic": [
    {
      "name": "datacenter",
      "value": "DC005"
    }
  ],
  "entity": {
    "id": "5e86ec0a",
    "href": "/resourceInventory/v4/resource/5e86ec0a",
    "name": "NetworkSliceSubnet=1,ManagedElement=1,AMFFunction=1",
    "@referredType": "ResourceFunction"
```

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```
    }
}
]
```

Retrieve vertex

GET graph/{graphId}/vertex/{id}?fields=...&{filtering}

Description

This operation retrieves a vertex entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a Vertex:

Request

```
GET /tmf-api/topologyDiscovery/v4graph/b965a5b8/vertex/8e516fb0
Accept: application/json
```

Response

```
200

{
  "id": "8e516fb0",
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
  "name": "AMF",
  "description": "5G Core AMF",
  "@type": "Vertex",
  "vertexSpecification": {
    "id": "6106de14",
    "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
    "name": "VNF",
    "@referredType": "VertexSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edge": [
```

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```
      {
        "id": "446d7249",
        "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
        "name": "N8",
        "@referredType": "Edge"
      }
    ],
    "vertexCharacteristic": [
      {
        "name": "datacenter",
        "value": "DC005"
      }
    ],
    "entity": {
      "id": "5e86ec0a",
      "href": "/resourceInventory/v4/resource/5e86ec0a",
      "name": "NetworkSliceSubnet=1,ManagedElement=1,AMFFunction=1",
      "@referredType": "ResourceFunction"
    }
}
```


Create vertex

POST graph/{graphId}/vertex

Description

This operation creates a vertex entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Vertex, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
graph	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
description	
edge	
entity	
subGraph	
vertexCharacteristic	
vertexSpecification	

Experimental

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Usage Samples

Here's an example of a request for creating a Vertex resource. In this example the request only passes mandatory attributes.

Request

```
POST /tmf-api/topologyDiscovery/v4/vertex
Content-Type: application/json

{
   "name": "AMF",
   "graph": {
      "id": "b965a5b8",
      "href": "/topologyDiscovery/v4/graph/b965a5b8",
      "name": "5GC",
      "@referredType": "Graph"
   }
}
```

Response

```
201

{
   "id": "8e516fb0",
   "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
   "name": "AMF",
   "graph": {
      "id": "b965a5b8",
      "href": "/topologyDiscovery/v4/graph/b965a5b8",
      "name": "5GC",
      "@referredType": "Graph"
   }
}
```

Patch vertex

`PATCH graph/{graphId}/vertex/{id}`

Description

This operation allows partial updates of a vertex entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

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Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
href	
id	
@baseType	
@schemaLocation	
@type	
description	
edge	
entity	
graph	
name	
subGraph	
vertexCharacteristic	
vertexSpecification	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a Vertex resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/vertex/8e516fb0
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "8e516fb0",
```

Experimental

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```
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
  "name": "new name",
  "description": "5G Core AMF",
  "@type": "Vertex",
  "vertexSpecification": {
    "id": "6106de14",
    "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
    "name": "VNF",
    "@referredType": "VertexSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edge": [
    {
      "id": "446d7249",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
      "name": "N8",
      "@referredType": "Edge"
    }
  ],
  "vertexCharacteristic": [
    {
      "name": "datacenter",
      "value": "DC005"
    }
  ],
  "entity": {
    "id": "5e86ec0a",
    "href": "/resourceInventory/v4/resource/5e86ec0a",
    "name": "NetworkSliceSubnet=1,ManagedElement=1,AMFFunction=1",
    "@referredType": "ResourceFunction"
  }
}
```


Delete vertex

DELETE graph/{graphId}/vertex/{id}

Description

This operation deletes a vertex entity.

Usage Samples

Here's an example of a request for deleting a Vertex resource.

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Request
DELETE /tmf-api/topologyDiscovery/v4/vertex/42
Response
204

Operations on Edge

List edges

GET graph/{graphId}/edge?fields=...&{filtering}

Description

This operation list edge entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving Edge resources.

Request
GET /tmf-api/topologyDiscovery/v4/edge Accept: application/json
Response

200
 	
[
{
"id": "446d7249",
"href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
"name": "N8",
"description": "5G Core N8 Reference Interface",
"@type": "Edge",
"edgeSpecification": {

Experimental

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```
      "id": "95e66eb7",
      "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
      "name": "Transport",
      "@referredType": "EdgeSpecification"
    },
    "graph": {
      "id": "b965a5b8",
      "href": "/topologyDiscovery/v4/graph/b965a5b8",
      "name": "5GC",
      "@referredType": "Graph"
    },
    "edgeCharacteristic": [
      {
        "name": "provider",
        "value": "SDN45"
      }
    ],
    "bidirectional": true,
    "vertex": [
      {
        "id": "8e516fb0",
        "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
        "name": "AMF",
        "@referredType": "Vertex"
      },
      {
        "id": "9b9f1c66",
        "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/9b9f1c66",
        "name": "UDM",
        "@referredType": "Vertex"
      }
    ],
    "entity": {
      "id": "1fbfac41",
      "href": "/resourceInventory/v4/resource/1fbfac41",
      "name": "NetworkSlice=1,NetworkSliceSubnet=1,ManagedElement=1,Link_AMF_UDM=1",
      "@referredType": "ResourceFunction"
    }
  }
]
```


Retrieve edge

`GET graph/{graphId}/edge/{id}?fields=...&{filtering}`

Description

This operation retrieves an edge entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

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Usage Samples

The following is an example of a request to get an Edge:

Request

```
GET /tmf-api/topologyDiscovery/v4graph/b965a5b8/edge/446d7249
Accept: application/json
```

Response

```
200

{
  "id": "446d7249",
  "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
  "name": "N8",
  "description": "5G Core N8 Reference Interface",
  "@type": "Edge",
  "edgeSpecification": {
    "id": "95e66eb7",
    "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
    "name": "Transport",
    "@referredType": "EdgeSpecification"
  },
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edgeCharacteristic": [
    {
      "name": "provider",
      "value": "SDN45"
    }
  ],
  "bidirectional": true,
  "vertex": [
    {
      "id": "8e516fb0",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
      "name": "AMF",
      "@referredType": "Vertex"
    },
    {
      "id": "9b9f1c66",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/9b9f1c66",
      "name": "UDM",
      "@referredType": "Vertex"
    }
```

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```
  ],
  "entity": {
    "id": "1fbfac41",
    "href": "/resourceInventory/v4/resource/1fbfac41",
    "name": "NetworkSlice=1,NetworkSliceSubnet=1,ManagedElement=1,Link_AMF_UDM=1",
    "@referredType": "ResourceFunction"
  }
}
```


Create edge

POST graph/{graphId}/edge

Description

This operation creates an edge entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Edge, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
graph	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
bidirectional	
description	
edgeCharacteristic	
edgeSpecification	
entity	
subGraph	
vertex	

Usage Samples

Here's an example of a request for creating a Edge resource. In this example the request only passes mandatory attributes.

Request

Experimental

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```
POST /tmf-api/topologyDiscovery/v4/edge
Content-Type: application/json

{
   "name": "N8",
   "graph": {
      "id": "b965a5b8",
      "href": "/topologyDiscovery/v4/graph/b965a5b8",
      "name": "5GC",
      "@referredType": "Graph"
   }
}
```

Response

```
201

{
   "id": "446d7249",
   "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
   "name": "N8",
   "graph": {
      "id": "b965a5b8",
      "href": "/topologyDiscovery/v4/graph/b965a5b8",
      "name": "5GC",
      "@referredType": "Graph"
   }
}
```

Patch edge

`PATCH graph/{graphId}/edge/{id}`

Description

This operation allows partial updates of an edge entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule

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href	
id	
@baseType	
@schemaLocation	
@type	
bidirectional	
description	
edgeCharacteristic	
edgeSpecification	
entity	
graph	
name	
subGraph	
vertex	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a Edge resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/edge/446d7249
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "446d7249",
   "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/446d7249",
   "name": "new name",
   "description": "5G Core N8 Reference Interface",
   "@type": "Edge",
   "edgeSpecification": {
      "id": "95e66eb7",
      "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
      "name": "Transport",
      "@referredType": "EdgeSpecification"
   },
```

Experimental

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```
  "graph": {
    "id": "b965a5b8",
    "href": "/topologyDiscovery/v4/graph/b965a5b8",
    "name": "5GC",
    "@referredType": "Graph"
  },
  "edgeCharacteristic": [
    {
      "name": "provider",
      "value": "SDN45"
    }
  ],
  "bidirectional": true,
  "vertex": [
    {
      "id": "8e516fb0",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/8e516fb0",
      "name": "AMF",
      "@referredType": "Vertex"
    },
    {
      "id": "9b9f1c66",
      "href": "/topologyDiscovery/v4/graph/b965a5b8/vertex/9b9f1c66",
      "name": "UDM",
      "@referredType": "Vertex"
    }
  ],
  "entity": {
    "id": "1fbfac41",
    "href": "/resourceInventory/v4/resource/1fbfac41",
    "name": "NetworkSlice=1,NetworkSliceSubnet=1,ManagedElement=1,Link_AMF_UDM=1",
    "@referredType": "ResourceFunction"
  }
}
```


Delete edge

`DELETE graph/{graphId}/edge/{id}`

Description

This operation deletes an edge entity.

Usage Samples

Here's an example of a request for deleting a Edge resource.

Request

Exp

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DELETE /tmf-api/topologyDiscovery/v4/edge/42
Response
204

Operations on Vertex Specification

List vertex specifications

`GET /vertexSpecification?fields=...&{filtering}`

Description

This operation list vertex specification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving VertexSpecification resources.

Request
GET /tmf-api/topologyDiscovery/v4/vertexSpecification Accept: application/json
Response
200 [{ "id": "6106de14", "href": "/topologyDiscovery/v4/vertexSpecification/6106de14", "@type": "VertexSpecification", "name": "VNF", "description": "Virtual Network Function (VNF)", "lastUpdate": "2021-04-08T12:22:08Z", "lifecycleStatus": "Active", "version": "1", "vertexSpecCharacteristic": [

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```
    {
      "name": "datacenter",
      "description": "Identifies the datacenter where the VNF is located",
      "valueType": "String",
      "configurable": true
    }
  ]
}
]
```

Retrieve vertex specification

`GET /vertexSpecification/{id}?fields=...&{filtering}`

Description

This operation retrieves a vertex specification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a VertexSpecification:

Request

```
GET /tmf-api/topologyDiscovery/v4/vertexSpecification/6106de14
Accept: application/json
```

Response

```
200

{
  "id": "6106de14",
  "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
  "@type": "VertexSpecification",
  "name": "VNF",
  "description": "Virtual Network Function (VNF)",
  "lastUpdate": "2021-04-08T12:22:08Z",
  "lifecycleStatus": "Active",
  "version": "1",
  "vertexSpecCharacteristic": [
    {
      "name": "datacenter",
      "description": "Identifies the datacenter where the VNF is located",
```

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```
        "valueType": "String",
        "configurable": true
      }
   ]
}
```

Create vertex specification

POST /vertexSpecification

Description

This operation creates a vertex specification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a VertexSpecification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
description	
lastUpdate	
lifecycleStatus	
relatedParty	
validFor	
version	
vertexSpecCharacteristic	

Usage Samples

Here's an example of a request for creating a VertexSpecification resource. In this example the request only passes mandatory attributes.

Request

```
POST /tmf-api/topologyDiscovery/v4/vertexSpecification
Content-Type: application/json

{
```

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```
  "name": "VNF"
}
```


Response


```
201

{
  "id": "6106de14",
  "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
  "name": "VNF"
}
```


Patch vertex specification

PATCH /vertexSpecification/{id}

Description

This operation allows partial updates of a vertex specification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
href	
id	
description	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
validFor	
version	
vertexSpecCharacteristic	

Experimental

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Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a VertexSpecification resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/vertexSpecification/6106de14
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "6106de14",
   "href": "/topologyDiscovery/v4/vertexSpecification/6106de14",
   "@type": "VertexSpecification",
   "name": "new name",
   "description": "Virtual Network Function (VNF)",
   "lastUpdate": "2021-04-08T12:22:08Z",
   "lifecycleStatus": "Active",
   "version": "1",
   "vertexSpecCharacteristic": [
      {
         "name": "datacenter",
         "description": "Identifies the datacenter where the VNF is located",
         "valueType": "String",
         "configurable": true
      }
   ]
}
```

Delete vertex specification

`DELETE /vertexSpecification/{id}`

Description

This operation deletes a vertex specification entity.

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Usage Samples

Here's an example of a request for deleting a VertexSpecification resource.

Request
DELETE /tmf-api/topologyDiscovery/v4/vertexSpecification/42
Response
204

Operations on Edge Specification

List edge specifications

`GET /edgeSpecification?fields=...&{filtering}`

Description

This operation list edge specification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving EdgeSpecification resources.

Request
GET /tmf-api/topologyDiscovery/v4/edgeSpecification Accept: application/json
Response
200 [

Experimental

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```
{
  "id": "95e66eb7",
  "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
  "@type": "EdgeSpecification",
  "name": "Transport",
  "description": "Link transport domain",
  "lastUpdate": "2021-04-08T13:54:32Z",
  "lifecycleStatus": "Active",
  "version": "1",
  "edgeSpecCharacteristic": [
    {
      "name": "provider",
      "description": "Identifies the transport provider of the link",
      "valueType": "String",
      "configurable": true
    }
  ]
}
]
```


Retrieve edge specification

`GET /edgeSpecification/{id}?fields=...&{filtering}`

Description

This operation retrieves an edge specification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get an EdgeSpecification:

Request
GET /tmf-api/topologyDiscovery/v4/edgeSpecification/95e66eb7 Accept: application/json
Response
200 { "id": "95e66eb7", "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",

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```
  "@type": "EdgeSpecification",
  "name": "Transport",
  "description": "Link transport domain",
  "lastUpdate": "2021-04-08T13:54:32Z",
  "lifecycleStatus": "Active",
  "version": "1",
  "edgeSpecCharacteristic": [
    {
      "name": "provider",
      "description": "Identifies the transport provider of the link",
      "valueType": "String",
      "configurable": true
    }
  ]
}
```


Create edge specification

POST /edgeSpecification

Description

This operation creates an edge specification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a EdgeSpecification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
description	
edgeSpecCharacteristic	
lastUpdate	
lifecycleStatus	
relatedParty	
validFor	
version	

Usage Samples

Experimental

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Here's an example of a request for creating a EdgeSpecification resource. In this example the request only passes mandatory attributes.

Request

```
POST /tmf-api/topologyDiscovery/v4/edgeSpecification
Content-Type: application/json

{
   "name": "Transport"
}
```

Response

```
201

{
   "id": "95e66eb7",
   "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
   "name": "Transport"
}
```

Patch edge specification

`PATCH /edgeSpecification/{id}`

Description

This operation allows partial updates of an edge specification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
href	

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id	
description	
edgeSpecCharacteristic	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
validFor	
version	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a EdgeSpecification resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/edgeSpecification/95e66eb7
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "95e66eb7",
   "href": "/topologyDiscovery/v4/edgeSpecification/95e66eb7",
   "@type": "EdgeSpecification",
   "name": "new name",
   "description": "Link transport domain",
   "lastUpdate": "2021-04-08T13:54:32Z",
   "lifecycleStatus": "Active",
   "version": "1",
   "edgeSpecCharacteristic": [
      {
         "name": "provider",
         "description": "Identifies the transport provider of the link",
         "valueType": "String",
         "configurable": true
      }
   ]
}
```

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Delete edge specification

```
DELETE /edgeSpecification/{id}
```

Description

This operation deletes an edge specification entity.

Usage Samples

Here's an example of a request for deleting a EdgeSpecification resource.

Request
DELETE /tmf-api/topologyDiscovery/v4/edgeSpecification/42
Response
204

Operations on Topology Discovery Job

List topology discovery jobs

```
GET /topologyDiscoveryJob?fields=...&{filtering}
```

Description

This operation list topology discovery job entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving TopologyDiscoveryJob resources.

Request

Experimental

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GET /tmf-api/topologyDiscovery/v4/topologyDiscoveryJob
Accept: application/json

Response

200


```
[
{
   "id": "b32a6d3b",
   "href": "/topologyDiscovery/v4/topologyDiscoveryJob/b32a6d3b",
   "@type": "TopologyDiscoveryJob",
   "creationDate": "2021-04-15T14:17:04.908Z",
   "completionDate": "2021-04-15T14:17:11.845Z",
   "status": "done",
   "topologyProvider": [
      {
         "url": "/resourceInventoryManagement/v4/resource?category=5GC"
      }
   ],
   "graphTemplate": {
      "name": "5GC",
      "description": "5G Core Subnet",
      "contained": true
   },
   "graphResult": {
      "id": "41fbbd95",
      "href": "/topologyDiscovery/v4/graph/41fbbd95",
      "@referredType": "Graph"
   }
}
]
```


Retrieve topology discovery job

`GET /topologyDiscoveryJob/{id}?fields=...&{filtering}`

Description

This operation retrieves a topology discovery job entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a TopologyDiscoveryJob:

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Request

```
GET /tmf-api/topologyDiscovery/v4/topologyDiscoveryJob/b32a6d3b
Accept: application/json
```

Response

```
200

{
  "id": "8917e8d0",
  "href": "/topologyDiscovery/v4/topologyDiscoveryJob/8917e8d0",
  "@type": "TopologyDiscoveryJob",
  "creationDate": "2021-04-15T14:17:04.908Z",
  "completionDate": "2021-04-15T14:17:11.845Z",
  "status": "done",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ],
  "graphResult": {
    "id": "aadd6dee",
    "href": "/topologyDiscovery/v4/graph/aadd6dee",
    "@type": "Graph"
  }
}
```

Create topology discovery job

`POST /topologyDiscoveryJob`

Description

This operation creates a topology discovery job entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a TopologyDiscoveryJob, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
topologyProvider	

Non Mandatory Attributes	Rule

Expert

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@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
status	

Usage Samples

The following is an example of an asynchronous request to start a TopologyDiscoveryJob. The topology provider is a resource inventory (TMF639) and the query filters for just the inventory items with category 5GC.

Request

```
POST /tmf-api/topologyDiscovery/v4/topologyDiscoveryJob
Content-Type: application/json

{
  "@type": "TopologyDiscoveryJob",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ]
}
```

Response

```
201

{
  "id": "8917e8d0",
  "href": "/topologyDiscovery/v4/topologyDiscoveryJob/8917e8d0",
  "@type": "TopologyDiscoveryJob",
  "creationDate": "2021-04-15T14:17:04.908Z",
  "status": "accepted",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ]
}
```

The following is an example of a synchronous request for a TopologyDiscoveryJob. The topology provider is a product catalog (TMF620) and a single product offering is requested. The response is a graph created with the

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default discovery rules which respresents the relationships as edges and the referenced items as vertices. Edges take a name value of the attribute of the relationship and vertex names mirror the name of the entity.

Request

```
POST /tmf-api/topologyDiscovery/v4/topologyDiscoveryJob
Content-Type: application/json

{
  "@type": "TopologyDiscoveryJob",
  "topologyProvider": [
    {
      "url": "/productCatalogManagement/v4/productOffering/7655"
    }
  ]
}
```

Response

```
201

{
  "id": "83016664",
  "href": "/topologyDiscovery/v4/topologyDiscoveryJob/83016664",
  "@type": "TopologyDiscoveryJob",
  "creationDate": "2021-04-15T17:03:46.234Z",
  "completionDate": "2021-04-15T17:03:47.807Z",
  "status": "done",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ],
  "graphResult": {
    "id": "fcf09cb3",
    "href": "/topologyDiscovery/v4/graph/fcf09cb3",
    "@type": "Graph"
  }
}
```

Patch topology discovery job

`PATCH /topologyDiscoveryJob/{id}`

Description

This operation allows partial updates of a topology discovery job entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is

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optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
href	
id	
@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
status	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a TopologyDiscoveryJob resource.

Request

```
PATCH /tmf-api/topologyDiscovery/v4/topologyDiscoveryJob/b32a6d3b
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "b32a6d3b",
   "href": "/topologyDiscovery/v4/topologyDiscoveryJob/b32a6d3b",
   "@type": "TopologyDiscoveryJob",
   "creationDate": "2021-04-15T14:17:04.908Z",
```

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```
  "completionDate": "2021-04-15T14:17:11.845Z",
  "status": "done",
  "topologyProvider": [
    {
      "url": "/resourceInventoryManagement/v4/resource?category=5GC"
    }
  ],
  "graphTemplate": {
    "name": "5GC",
    "description": "5G Core Subnet",
    "contained": true
  },
  "graphResult": {
    "id": "41fbbd95",
    "href": "/topologyDiscovery/v4/graph/41fbbd95",
    "@referredType": "Graph"
  },
  "name": "new name"
}
```


Operations on Merge Topology Graph Job

List merge topology graph jobs

`GET /mergeTopologyGraphJob?fields=...&{filtering}`

Description

This operation list merge topology graph job entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving MergeTopologyGraphJob resources.

Request
GET /tmf-api/topologyDiscovery/v4/mergeTopologyGraphJob Accept: application/json
Response
200 [

Experimental

```
{
  "id": "333c231a",
  "href": "/topologyDiscovery/v4/MergeTopologyGraphJob/333c231a",
  "@type": "MergeTopologyGraphJob",
  "creationDate": "2021-04-15T14:37:18.762Z",
  "completionDate": "2021-04-15T14:37:19.114Z",
  "status": "done",
  "mergeTopologyGraph": [
    {
      "id": "91f9e74a",
      "href": "/topologyDiscovery/v4/graph/91f9e74a",
      "@referredType": "Graph"
    },
    {
      "id": "0319108c",
      "href": "/topologyDiscovery/v4/graph/0319108c",
      "@referredType": "Graph"
    }
  ],
  "graph": {
    "id": "ded3c8be",
    "href": "/topologyDiscovery/v4/graph/ded3c8be",
    "@referredType": "Graph"
  }
}
]
```


Retrieve merge topology graph job

`GET /mergeTopologyGraphJob/{id}?fields=...&{filtering}`

Description

This operation retrieves a merge topology graph job entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a MergeTopologyGraphJob:

Request
GET /tmf-api/topologyDiscovery/v4/mergeTopologyGraphJob/84ed0779 Accept: application/json

Experimental

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Response

200
 	
{
"id": "84ed0779",
"href": "/topologyDiscovery/v4/MergeTopologyGraphJob/84ed0779",
"@type": "MergeTopologyGraphJob",
"creationDate": "2021-04-15T14:17:04.908Z",
"status": "inProgress",
"mergeTopologyGraph": [
{
"id": "2ffe2827",
"href": "/topologyDiscovery/v4/graph/2ffe2827",
"@referredType": "Graph"
},
{
"id": "0a790d6c",
"href": "/topologyDiscovery/v4/graph/0a790d6c",
"@referredType": "Graph"
}
]
}

Create merge topology graph job

`POST /mergeTopologyGraphJob`

Description

This operation creates a merge topology graph job entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a MergeTopologyGraphJob, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
mergeTopologyGraph	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
graphResult	
status	

Experimental

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Usage Samples

The following is an example of an asynchronous request to start a MergeTopologyGraphJob:

Request

```
POST /tmf-api/topologyDiscovery/v4/mergeTopologyGraphJob
Content-Type: application/json

{
  "@type": "MergeTopologyGraphJob",
  "mergeTopologyGraph": [
    {
      "id": "2ffe2827",
      "href": "/topologyDiscovery/v4/graph/2ffe2827",
      "@referredType": "Graph"
    },
    {
      "id": "0a790d6c",
      "href": "/topologyDiscovery/v4/graph/0a790d6c",
      "@referredType": "Graph"
    }
  ]
}
```

Response

```
201

{
  "id": "84ed0779",
  "href": "/topologyDiscovery/v4/MergeTopologyGraphJob/84ed0779",
  "@type": "MergeTopologyGraphJob",
  "creationDate": "2021-04-15T14:17:04.908Z",
  "completionDate": "2021-04-15T14:17:11.845Z",
  "status": "done",
  "mergeTopologyGraph": [
    {
      "id": "2ffe2827",
      "href": "/topologyDiscovery/v4/graph/2ffe2827",
      "@referredType": "Graph"
    },
    {
      "id": "0a790d6c",
      "href": "/topologyDiscovery/v4/graph/0a790d6c",
      "@referredType": "Graph"
    }
  ],
```

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```
  "graphResult": {
    "id": "39c6230e",
    "href": "/topologyDiscovery/v4/graph/39c6230e",
    "@referredType": "Graph"
  }
}
```

Patch merge topology graph job

PATCH /mergeTopologyGraphJob/{id}

Description

This operation allows partial updates of a merge topology graph job entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
href	
id	
@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
graphResult	
mergeTopologyGraph	
status	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a MergeTopologyGraphJob resource.

Request

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PATCH /tmf-api/topologyDiscovery/v4/mergeTopologyGraphJob/333c231a
Content-Type: application/merge-patch+json

```
{
  "name": "new name"
}
```

Response

200


```
{
  "id": "333c231a",
  "href": "/topologyDiscovery/v4/MergeTopologyGraphJob/333c231a",
  "@type": "MergeTopologyGraphJob",
  "creationDate": "2021-04-15T14:37:18.762Z",
  "completionDate": "2021-04-15T14:37:19.114Z",
  "status": "done",
  "mergeTopologyGraph": [
    {
      "id": "91f9e74a",
      "href": "/topologyDiscovery/v4/graph/91f9e74a",
      "@referredType": "Graph"
    },
    {
      "id": "0319108c",
      "href": "/topologyDiscovery/v4/graph/0319108c",
      "@referredType": "Graph"
    }
  ],
  "graph": {
    "id": "ded3c8be",
    "href": "/topologyDiscovery/v4/graph/ded3c8be",
    "@referredType": "Graph"
  },
  "name": "new name"
}
```


Operations on Topology Query Job

List topology query jobs

`GET /topologyQueryJob?fields=...&{filtering}`

Description

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This operation list topology query job entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving TopologyQueryJob resources.

Request

```
GET /tmf-api/topologyDiscovery/v4/topologyQueryJob
Accept: application/json
```

Response

```
200

[
{
   "id": "714733ce",
   "href": "/topologyDiscovery/v4/topologyQueryJob/714733ce",
   "@type": "TopologyQueryJob",
   "creationDate": "2021-04-16T33:21:54.776Z",
   "completionDate": "2021-04-16T33:21:55.245Z",
   "status": "done",
   "query": "TUFUQ0ggKG4pClJFVFVSTiBu",
   "graphTemplate": {
      "description": "Query result subgraph",
      "contained": false
   },
   "graphResult": {
      "id": "c765bfa9",
      "href": "/topologyDiscovery/v4/graph/c765bfa9",
      "@referredType": "Graph"
   }
}
]
```

Retrieve topology query job

`GET /topologyQueryJob/{id}?fields=...&{filtering}`

Description

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This operation retrieves a topology query job entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a TopologyQueryJob:

Request

```
GET /tmf-api/topologyDiscovery/v4/topologyQueryJob/c765bfa9
Accept: application/json
```

Response

```
200

{
   "id": "c765bfa9",
   "href": "/topologyDiscovery/v4/topologyQueryJob/c765bfa9",
   "@type": "TopologyQueryJob",
   "creationDate": "2021-04-16T47:45:15.023Z",
   "completionDate": "2021-04-16T47:45:16.144Z",
   "status": "done",
   "graphSource": {
      "id": "451f36cc",
      "href": "/topologyDiscovery/v4/graph/451f36cc",
      "@type": "Graph"
   },
   "query": "TUFUQ0ggKG4pClJFVFVSTiBu",
   "graphResult": {
      "id": "ede7e125",
      "href": "/topologyDiscovery/v4/graph/ede7e125",
      "@type": "Graph"
   }
}
```

Create topology query job

`POST /topologyQueryJob`

Description

This operation creates a topology query job entity.

Mandatory and Non Mandatory Attributes

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The following tables provide the list of mandatory and non mandatory attributes when creating a TopologyQueryJob, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
graphSource	
query	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
graphResult	
graphTemplate	
queryMatch	
status	
vertexResult	

Usage Samples

The following is an example of an asynchronous request to start a TopologyQueryJob.

Request

```
POST /tmf-api/topologyDiscovery/v4/topologyQueryJob
Content-Type: application/json

{
   "@type": "TopologyQueryJob",
   "graphSource": {
      "id": "451f36cc",
      "href": "/topologyDiscovery/v4/graph/451f36cc",
      "@type": "Graph"
   },
   "query": "TUFUQ0ggKG4pClJFVFVSTiBu"
}
```

Response

```
201

{
   "id": "c765bfa9",
   "href": "/topologyDiscovery/v4/topologyQueryJob/c765bfa9",
```

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```
  "@type": "TopologyQueryJob",
  "creationDate": "2021-04-16T47:45:15.023Z",
  "status": "accepted",
  "graphSource": {
    "id": "451f36cc",
    "href": "/topologyDiscovery/v4/graph/451f36cc",
    "@type": "Graph"
  },
  "query": "TUFUQ0ggKG4pClJFVFVSTiBu"
}
```


The following is an example of a synchronous request for a TopologyQueryJob.

Request

```
POST /tmf-api/topologyDiscovery/v4/topologyQueryJob
Content-Type: application/json

{
  "@type": "TopologyQueryJob",
  "graphSource": {
    "id": "37e9754d",
    "href": "/topologyDiscovery/v4/graph/37e9754d",
    "@type": "Graph"
  },
  "query": "TUFUQ0ggKG4pClJFVFVSTiBu"
}
```

Response

```
201

{
  "id": "83016664",
  "href": "/topologyDiscovery/v4/topologyDiscoveryJob/83016664",
  "@type": "TopologyDiscoveryJob",
  "creationDate": "2021-04-15T18:12:59.203Z",
  "status": "done",
  "graphSource": {
    "id": "37e9754d",
    "href": "/topologyDiscovery/v4/graph/37e9754d",
    "@type": "Graph"
  },
  "query": "TUFUQ0ggKG4pClJFVFVSTiBu",
  "graphResult": {
    "id": "34608ba9",
    "href": "/topologyDiscovery/v4/graph/34608ba9",
    "@type": "Graph"
  }
```

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```
}
```

Patch topology query job

PATCH /topologyQueryJob/{id}

Description

This operation allows partial updates of a topology query job entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
href	
id	
@baseType	
@schemaLocation	
@type	
completionDate	
creationDate	
graphResult	
graphSource	
graphTemplate	
queryMatch	
status	
vertexResult	

Non Patchable Attributes	Rule

Usage Samples

Here's an example of a request for patching a TopologyQueryJob resource.

Request
PATCH /tmf-api/topologyDiscovery/v4/topologyQueryJob/714733ce

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Content-Type: application/merge-patch+json


```
{
   "name": "new name"
}
```


Response

200


```
{
   "id": "714733ce",
   "href": "/topologyDiscovery/v4/topologyQueryJob/714733ce",
   "@type": "TopologyQueryJob",
   "creationDate": "2021-04-16T33:21:54.776Z",
   "completionDate": "2021-04-16T33:21:55.245Z",
   "status": "done",
   "query": "TUFUQ0ggKG4pClJFVFVSTiBu",
   "graphTemplate": {
      "description": "Query result subgraph",
      "contained": false
   },
   "graphResult": {
      "id": "c765bfa9",
      "href": "/topologyDiscovery/v4/graph/c765bfa9",
      "@referredType": "Graph"
   },
   "name": "new name"
}
```


Expe

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

`POST /hub`

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request
POST /api/hub Accept: application/json {"callback": "http://in.listener.com"}
Response
201 Content-Type: application/json Location: /api/hub/42 {"id":"42","callback":"http://in.listener.com","query":null}

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request
DELETE /api/hub/42 Accept: application/json
Response
204

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

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Here's an example of a notification received by the listener. In this example “EVENT TYPE” should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

POST /client/listener
Accept: application/json
 	
{
"event": {
EVENT BODY
},
"eventType": "EVENT_TYPE"
}

Response
201

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

Experimental

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description

Contributors to Document

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